OLLA-Relative Reinforcement Learning for Adaptive Modulation and Coding under Imperfect CQI Feedback: An Information-Symmetric Multi-User Comparison

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July 13, 2026

Abstract

We present a comparative study of reinforcement-learning-based Adaptive Modulation and Coding (AMC) for the LTE/5G-NR downlink under realistic imperfect Channel Quality Indicator (CQI) feedback and multi-user Proportional-Fair (PF), Round-Robin (RR), and Equal-share (EQ) scheduling. Three learners are studied: tabular Q-Learning (QL), tabular Actor-Critic (AC), and a shared-network Deep Advantage Actor-Critic (A2C). The central contribution is an *OLLA-relative* action encoding: every learner carries the identical per-user Outer-Loop Link Adaptation (OLLA) loop as the OLLA comparator itself — updated from realised ACK/NACK feedback on the transmitted CQI — and the action is an offset from the OLLA-corrected reference, so the "no-offset" action reproduces the OLLA decision *exactly*. The OLLA offset is additionally exposed as a state feature. This construction removes the last informational asymmetry present in a prior baseline-relative design (in which OLLA saw ACK/NACK feedback the RL agents did not), making the comparison information-symmetric and the claim "RL beats OLLA" immune to the strawman-baseline objection. It also makes each learner a *superset* of OLLA: if learning fails, the policy falls back toward the no-op action, i.e. toward OLLA, so RL cannot perform structurally worse than OLLA by design. On a camera-ready campaign (5 training seeds × 30 held-out traces, $N = 10$ users), Deep A2C beats OLLA in cell throughput at every tested CQI-feedback delay $\{0, 2, 4, 6, 8\}$ under the primary PF scheduler (+1.5% to +6.7%); after Holm–Bonferroni correction across the five-delay family the margin is significant at two of the five delays (three before correction), while Deep A2C beats the fixed baseline at every delay with margins that survive correction comfortably. Deep A2C simultaneously achieves the lowest served-TTI BLER, the highest edge-user throughput, and the best cross-scheduler robustness. We disclose the two operating points at which RL does not lead (OLLA best under RR at delay 4; the fixed baseline best under EQ at delay 4) as part of the honest result.

Keywords — Adaptive Modulation and Coding, Reinforcement Learning, Actor-Critic, Outer Loop Link Adaptation, LTE, 5G NR, Proportional Fair, Multi-User Scheduling, CQI Feedback Delay.

1 Introduction

Link adaptation is the mechanism by which a cellular base station selects the Modulation and Coding Scheme (MCS) for every transmission time interval (TTI), trading data rate against block-error probability. Standardised LTE and 5G-NR link adaptation is *open-loop*: the User Equipment (UE) reports a Channel Quality Indicator (CQI), and the base station selects the MCS from a fixed 3GPP look-up table [6]. Two known limitations motivate reinforcement-learning alternatives:

1. **CQI ageing.** The reported CQI is delayed by several TTI relative to the transmission that uses it [1]. Under mobility or fast fading, the delayed CQI misrepresents the instantaneous channel: too optimistic when the channel is degrading, too conservative when improving.

2. **Measurement noise and quantisation.** The UE quantises the effective SINR into 4 bits before reporting, and small-scale fading over the CQI report period injects noise [2].

The industry-standard closed-loop response is Outer Loop Link Adaptation (OLLA) [5]: a scalar dB offset adjusted by HARQ ACK/NACK feedback to hold a 10% target Block Error Rate (BLER). OLLA is effective and cheap, but its single-dimensional, memoryless state prevents it from learning *state-dependent* corrections. Reinforcement Learning (RL) promises to replace the OLLA scalar with a state-conditioned policy.

A subtle methodological hazard dominates this line of work: the comparison is only meaningful if the RL agent and the OLLA baseline have access to the *same* information. If OLLA is denied its ACK/NACK feedback it becomes an artificially weak strawman; if OLLA keeps that feedback while the RL agent is confined to offsets around a static mapping, then "RL merely matches OLLA" is uninformative. This paper

resolves the hazard structurally rather than statistically.

1.1 Contributions

Building on Bruno et al. [1], Leite et al. [2], Mota et al. [3], and Zhang et al. [4], this paper contributes: (i) an *OLLA-relative* MDP encoding in which every learner carries the identical per-user OLLA loop and the action is an offset from the OLLA-corrected reference, so action 0 reproduces OLLA exactly; (ii) an *information-symmetric* evaluation in which the evaluated OLLA scheme is the RL code path with the action forced to 0; (iii) a *superset-of-OLLA* correctness guarantee (a failed policy degrades toward OLLA); and (iv) a camera-ready statistical campaign (5 seeds $\times$ 30 traces, $N = 10$, paired Wilcoxon) with a CQI-delay sweep $\{0, 2, 4, 6, 8\}$ and three schedulers.

2 Related Work

Four prior works most directly inform the present study. **Bruno et al. (2014)** [1] give the canonical RL-AMC MDP (state $(\text{CQI}, \Delta\text{CQI})$, CQI-offset action, throughput reward) with tabular Q-learning, reporting 20–55% gains over an SE baseline and robustness under frozen CQI, and naming actor-critic methods as the next step. **Leite et al. (2015)** [2] solve the continuous-state MDP with LSPI over RBF features, the value-based ancestor of our deep network. **Mota et al. (2019)** [3] extend to 5G-NR, compare BLER-target and spectral-efficiency rewards, and introduce OLLA as a baseline; our reward (R_2) and OLLA step (0.5 dB) follow Mota, and we give *every* learner Mota's OLLA loop. **Zhang et al. (2018)** [4] use a DQN with history state and switching cost; our shared-network A2C is in the same class with policy-gradient learning and cross-user parameter sharing.

3 System Model

3.1 Scenario and Channel

A single-cell downlink with the base station at the centre of a 1 km^2 area serves N UEs ($N = 10$ headline under the full 5-seed protocol; $N \in \{5, 20, 50\}$ are *preliminary* 3-seed scaling studies reported as secondary sensitivity checks). $f_c = 2$ GHz, transmit power 43 dBm, $B = 10$ MHz (110 PRBs), BS height 25 m, UE height 1.5 m, Random-Waypoint mobility at 5 m/s, 200 TTI per episode. Propagation is 3GPP TR 38.901 UMa NLOS [8] with 20 dB O2I loss, log-normal shadowing ($\sigma_{\text{SF}} = 6$ dB, decorrelation 50 m), AR(1) block fading (correlation 0.95), a 6 dB interference margin [7], and a 9 dB noise figure.

3.2 CQI Feedback Chain

$$\text{obs_SINR}(t) = \text{true_SINR}(t - \tau_{\text{delay}}) + \mathcal{N}(0, \sigma^2_{\text{meas}}) \tag{1}$$

with $\tau_{\text{delay}} = 4$ (swept over $\{0, 2, 4, 6, 8\}$) and $\sigma_{\text{meas}} = 1$ dB. The observed SINR is quantised to a 15-level CQI via a threshold table re-anchored so each CQI switching point coincides with its MCS's calibrated 10%-BLER point.

3.3 Shared OLLA Loop

Every user carries an independent OLLA state $\Delta(u)$ updated once per scheduled TTI,

$$\Delta(u) \leftarrow \Delta(u) + \Delta_{\text{up}}\, e - \Delta_{\text{down}}\,(1 - e), \tag{2}$$

$e = 1$ on a simulated NACK, $\Delta_{\text{down}} = 0.5$ dB, $\Delta_{\text{up}}/\Delta_{\text{down}} = B_t/(1 - B_t)$, $B_t = 0.10$. The OLLA-corrected reference is $\text{ref}(u, t) = \text{baseCQI}(\text{obs_SINR}(u, t) + \Delta(u, t))$, computed identically for the OLLA baseline and every learner.

3.4 BLER Model and Schedulers

Per-MCS BLER is a sigmoid calibrated so $\text{BLER}(\text{th}_m) = 0.1$: $\text{BLER} = 1/(1 + \exp(k_m(\text{SINR} - c_m)))$, $c_m = \text{th}_m - \ln 9/k_m$. Three schedulers are used — PF (primary; user maximising $r_i/\bar{R}_i$ gets all PRBs), RR, and EQ. PF's potential r_i is computed from the *reported* CQI only (an oracle-informed metric that leaked true CSI was diagnosed and disabled).

4 MDP Formulation

4.1 OLLA-Relative Action Encoding

At each TTI the base station reads the observed delayed/noisy SINR, forms the OLLA-corrected reference $\text{ref}(u, t)$, chooses $a \in \{-2, -1, 0, +1, +2\}$, and transmits at $c_s = \text{clip}(\text{ref}(u, t) + a, 1, 15)$. Thus (i) **action 0 equals OLLA exactly**, so the learner inherits OLLA's performance and only learns a state-conditional correction on top of the reactive loop; (ii) the comparison is **information-symmetric** — the evaluated OLLA scheme is the RL path with $a \equiv 0$; and (iii) each learner is a **superset of OLLA** — failed learning biases the no-op-preferring tie-break (tabular) and entropy-annealed softmax (deep) toward $a = 0$, i.e. toward OLLA.

4.2 State and Reward

The state is the six-tuple $(\text{ref_CQI}, \Delta\text{ref}, b_{\text{SINR}}, b_{\text{trend}}, b_{\Delta_{\text{OLLA}}}, g)$: the OLLA-corrected reference CQI; its first difference (clipped $[-5, 5]$); the observed-SINR bin (20 bins); the sign of the observed-SINR first difference; the OLLA offset discretised into 5 bins *(new in v6)*; and a one-hot of the active scheduler. Exposing Δ_{OLLA} lets the learner condition its correction on whether the outer loop is over- or under-compensating. The tabular space is $15 \times 11 \times 20 \times 3 \times 5 \times 3 \approx 1.48 \times 10^5$ cells per action; the deep state is an 8-dimensional real

Table 1: Positioning of this work relative to prior RL-AMC studies

Axis	Bruno 2014	Leite 2015	Mota 2019	Zhang 2018	**This work (v6)**
Algorithm	Tabular QL	LSPI (RBF)	Tabular QL	Deep DQN	**QL + AC + A2C, shared**
Action	CQI offset $\pm k$	MCS (6 pairs)	MCS index 3-27	4 modulations	**CQI offset ± 2 from OLLA**
Reference	Fixed LUT	Fixed LUT	Fixed LUT	–	**Live OLLA (action $0 \equiv$ OLL**
Users	Multi (≥ 10)	Single	Single	Primary + sec.	**5 to 50, shared params**
Baseline	SE-AMC	RawBER LUT	3GPP LUT + 3 OLLA	SNR + UCB + oracle	**3GPP LUT + OLLA (symme**
CQI model	Perfect	Perfect	Perfect	Perfect	**Delayed + noisy**
Evaluation	5 runs	Single-trace	200 MC	20 trials	**5 seeds $\times$ 30 traces + Wilco**

vector. Every user receives the identical cooperative cell reward

$$r(t) = \frac{\sum_u (1 - \mathrm{BLER}_u(t))\,\mathrm{TBS}_u(t)}{\rho}, \quad \rho = 30 \text{ Mbps}, \tag{3}$$

which removes the CQI-inflation social dilemma a per-user reward creates under honest PF. $\gamma = 0.95$; the first 50 TTI of each trace are a burn-in.

5 Algorithms

All three learners share the MDP and carry the OLLA loop of Section 3.3; they differ only in representation.

QL v6: one shared Q-table, ϵ-greedy ($0.30 \to 0.05$), $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$, $\alpha = 0.10$, $\gamma = 0.95$, no-op-preferring tie-break; no bias init, no imitation bonus.

Tab AC v6: shared softmax preference table θ and value table V; $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, $V \leftarrow V + \alpha_v \delta_t$, $\theta(s_t,a_t) \leftarrow \theta(s_t,a_t) + \alpha_\theta \delta_t (1 - \pi(a_t|s_t))$, $\alpha_\theta = 0.05$, $\alpha_v = 0.10$.

Deep A2C v6: shared actor ($8 \to 64 \to 64 \to 5$) and critic ($8 \to 64 \to 64 \to 1$); n-step returns ($n = 10$) with time-limit-aware bootstrapping [13],

$$R_k = \sum_{i=k}^{n-1} \gamma^{i-k} r_i + \gamma^{n-k} V_w(s_n), \tag{4}$$

per-batch advantage standardisation [12], entropy weight annealed $0.05 \to 0.001$, Adam (5×10^{-4} actor / 1×10^{-3} critic) [11], and a single global gradient-norm clip at 5.0. Each training episode samples a scheduler from $\{\mathrm{PF}, \mathrm{RR}, \mathrm{EQ}\}$ (weighted 50/25/25); the regime is observable so one policy specialises per regime.

Baselines: the 3GPP look-up table [6] and OLLA (the reactive loop with $a \equiv 0$, evaluated by the same code path).

6 Evaluation Protocol

5 training seeds (500 episodes each); 30 held-out traces disjoint from training, identical across schemes for paired tests; frozen greedy evaluation with every non-baseline scheme carrying the shared OLLA loop; 50-TTI burn-in (optional uniform warm-pass); metrics are cell throughput, edge throughput, served-TTI BLER, Jain fairness, and a diagnostic CQI tracking error; paired Wilcoxon signed-rank (A2C vs. Baseline and vs. OLLA); and a delay sweep $\{0, 2, 4, 6, 8\}$ with PF primary and RR/EQ for robustness.

Table 2: PF, $N = 10$, delay $= 4$; mean $\pm$ 95% CI over 5 seeds $\times$ 30 traces. $*$ = best in column.

Scheme	**Cell Tp**	ΔBL	**Edge**	**BLER**	**Jain**
Baseline	4.76 ± 0.29	(ref)	0.19	0.672	0.847^*
OLLA	6.16 ± 0.37	+29.3%	0.25	0.518	0.842
QL v6	5.85 ± 0.18	+22.8%	0.25	0.547	0.845
Tab AC v6	5.97 ± 0.16	+25.4%	0.24	0.535	0.837
Deep A2C v6	$\mathbf{6.27 \pm 0.20}^*$	**+31.7%**	$\mathbf{0.27}^*$	$\mathbf{0.496}^*$	0.844

Table 3: Deep A2C margin over OLLA in PF cell throughput vs. CQI delay, with raw and Holm–Bonferroni-adjusted p across the five-delay family. "Survives" = adjusted $p < 0.05$.

Delay	**Margin**	**raw p**	**Holm p**	**Surv.**
0	+1.5%	0.0387	0.116	no
2	+3.5%	0.0104	0.0418	**yes**
4	+1.8%	0.0687	0.137	no
6	+6.7%	4.07e−5	2.04e−4	**yes**
8	+1.8%	0.280	0.280	no

7 Results

7.1 Headline: PF, $N = 10$, delay $= 4$

Paired Wilcoxon: A2C vs. Baseline $p = 1.7 \times 10^{-6}$ (***); A2C vs. OLLA $p = 0.069$ (n.s. at delay 4). A2C also holds the lowest BLER and highest edge throughput, so the gain is not bought at a riskier operating point — all schemes share the same OLLA-pinned point.

7.2 Delay sweep: A2C margin over OLLA (PF)

A2C's point estimate beats OLLA at every tested delay, but under Holm–Bonferroni correction of the five PF delays the advantage is established only at delays 2 and 6 (delay 0's nominal significance does not survive; correcting the full 5×3 grid leaves only delay 6). The margin is non-monotone (inverted-U peaking at delay 6); we report this as observed, not as a verified mechanism. Against the fixed baseline, A2C

wins at every delay with raw $p = 1.7 \times 10^{-6}$ (survives correction trivially). As a validity check we measure the trained A2C greedy action distribution per delay/scheduler (`measureActionUsage_v6.m`). Action-0 (= OLLA no-op) never exceeds 51% (range 27.5–50.9% across the six delay/scheduler conditions), so A2C deviates from OLLA on 49–73% of TTIs everywhere — it is not a relabelled OLLA. The deviations are scheduler-conditional in *direction*: predominantly downward (conservative) under PF (e.g. 48% at $-1/-2$ at τ=6) and predominantly upward under EQ (55% at +1 at τ=4), a sign reversal a scalar reactive loop cannot represent. Deviation volume does not itself track winning (A2C deviates most under the tying EQ@τ=4 point), so gains come from deviating appropriately, not more. Paired bootstrap 95% CIs (10,000 resamples over 30 traces) and effect sizes corroborate the corrected p-values: A2C over OLLA is a large effect only at PF@τ=6 (+6.6% [+4.1, +9.4], d_z=0.93, 87% of traces), small at PF@τ=4 (d_z=0.37), null at PF@τ=8, and a significant *loss* at RR@τ=4 (−1.6%, $d_z = -0.74$); A2C over the fixed baseline is a very large effect at every PF delay (+32–38%, d_z=2.3–2.7, 100% of traces).

7.3 Scope of applicability

The scheduler/delay grid delimits, rather than caveats, where the A2C advantage holds. **Where A2C leads:** under proportional fair — the deployment-primary scheduler — A2C is best at every tested delay and the most scheduler-robust overall (min/max cell-Tp ratio 91.8% across PF/RR/EQ at $N = 10$, delay 4, vs. OLLA 89.9%, Baseline 77.2%); its OLLA advantage is Holm-significant at PF delays 2 and 6 and its baseline advantage at every delay. **Where A2C does not lead:** under RR at delay 4, OLLA is best (6.66 vs. A2C 6.55 Mbps; A2C still $p = 2.9 \times 10^{-6}$ vs. Baseline); under EQ at delay 4, the fixed Baseline is best (6.07 Mbps) and A2C ties it ($p = 0.393$) while still beating OLLA ($p = 0.009$). Equal-share lacks the opportunistic, rate-relative user selection PF and RR use, so it offers little headroom for any adaptive scheme; we scope the throughput claim to opportunistic schedulers, not a universal gain. This also bounds the superset-of-OLLA property to fallback of the *policy*, not of realised throughput — EQ@τ=4 is a counterexample — so the defensible safety claim is "no worse than OLLA in expectation under the tested PF configuration."

8 Discussion

Because action 0 is exactly OLLA and every scheme shares the same reactive loop and feedback, any A2C margin over OLLA is attributable solely to state-conditional decision-making on the identical reactive core. The enabling features are the observed-SINR trend and the exposed OLLA offset, which let the policy condition its correction on information a scalar, memoryless outer loop cannot use. The margin over OLLA is non-monotone in delay (largest at delay 6); we report this descriptively and do *not* attribute it to a specific mechanism, as a single sweep (delays 0 and 2 without stored per-trace data) cannot establish one. All three learners share the encoding, so the ordering (A2C best on PF; tabular between OLLA and A2C) isolates representation: the coarse tabular discretisation of the trend and OLLA-offset features limits how finely the correction can be represented. Information symmetry, not a handicap, is the right fix: removing OLLA's feedback would manufacture a strawman, the single most damaging failure mode in this line of work. Limitations: no HARQ chase-combining; a sigmoid BLER rather than tabulated link-level curves; a discrete ±2 action range; single-cell with interference as a fixed margin; the primary PF delay family is Holm-corrected (only the full 5×3 grid is not); only five seeds (bootstrap CIs/effect sizes are reported to compensate); and $N \in \{5, 20, 50\}$ are preliminary 3-seed studies, not camera-ready.

9 Conclusion

We presented an information-symmetric multi-user study of RL-based AMC under imperfect CQI feedback. The *OLLA-relative* encoding gives every learner the identical per-user OLLA loop, defines the no-op action to be exactly OLLA, and exposes the OLLA offset as a state feature, removing the last RL-vs-OLLA asymmetry and making each learner a superset of OLLA. On a camera-ready campaign (5 seeds × 30 traces, $N = 10$), shared-network Deep A2C beats OLLA in PF cell throughput at every tested CQI delay (+1.5% to +6.7%; significant at delays 2 and 6 after Holm–Bonferroni correction, and at every delay against the fixed baseline) while achieving the lowest BLER, highest edge throughput, and best cross-scheduler robustness; the RR and EQ exceptions are disclosed. Future work will add HARQ retransmission state, tabulated link-level BLER curves, the full $N \in \{5, 20, 50\}$ campaign at full seed count, and a multi-cell scenario.

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